CIVIL ENGINEERING

III B.Tech – I Semester

Course Code		L	T	P	C
23A01503P	**GEOTECHNICAL ENGINEERING LAB**	0	0	0	1.5

Course Objectives:

The objectives of this course are to make the student to:

1. **Understand** the fundamental index properties of soils and their significancein geotechnical engineering.
2. **Perform** field and laboratory tests to determine in-situ density and compaction characteristics of soils.
3. **Evaluate** the engineering properties of soil, including permeability, shear strength, and consolidation.
4. **Analyze** the strength and deformation characteristics of soils through shear and compression tests.
5. Interpret test results and relate engineering properties of soils to real-world geotechnical problems and design considerations.

Upon successful completion of this course, students will be able to:

1. **Determine** index properties of soil, including specific gravity, grain size distribution, and consistency limits.
2. **Conduct** field and laboratory compaction tests to evaluate the moisture-density relationship of soil.
3. **Evaluate** permeability and consolidation characteristics of soil using appropriate laboratory techniques.
4. **Analyze** the shear strength and compressibility of soil through direct shear, unconfined compression, and tri-axial tests.
5. **Integrate test results and engineering judgment** to interpret soil behavior and make informed decisionsin geotechnical engineering applications.

CO – PO Articulation Matrix:-

Course Outcomes	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO1**	**PSO2**
CO -1	3	2	2	2	-	-	-	-	-	2	-	2	3	2
CO -2	3	3	3	2	2	-	-	-	-	-	-	1	3	2
CO -3	3	3	2	2	2	-	-	-	-	-	-	1	3	3
CO -4	3	3	3	2	2	-	-	-	-	-	-	1	3	3
CO -5	2	2	2	1	3	-	-	-	-	2	1	2	2	2

LIST OF EXPERIMENTS :-

1. Determination of Index Properties
    a. Specific Gravity of Soil
    b. Grain Size Distribution – Sieve Analysis
    c. Grain Size Distribution - Hydrometer Analysis
    d. Liquid Limit and Plastic Limit Tests
    e. Shrinkage Limit and Differential Free Swell Tests
2. Determination of In-Situ Density and Compaction Characteristics
    a. Field Density Test (Sand Replacement Method)

b. Determination of Moisture–Density Relationship Using Standard Proctor Compaction Test.

3. Determination of Engineering Properties
   a. Permeability Determination (Constant Head Method)
   b. Permeability Determination (Falling Head Methods)
   c. Determination of Co-Efficient of Consolidation
   d. Direct Shear Test in Cohesion Less Soil
   e. Unconfined Compression Test in Cohesive Soil
   f. Laboratory Vane Shear Test in Cohesive Soil
   g. Tri-Axial Compression Test in Cohesion Less Soil
   h. California Bearing Ratio Test

Note: Any 10 of the above Experiments.

TEXT BOOKS:

1. Lambe T.W., "Soil Testing for Engineers", John Wiley and Sons, New York, 1951. Digitized 2008.
2. IS Code of Practice (2720) Relevant Parts, as amended from time to time, Bureau of Indian Standards, New Delhi.

REFRENCE BOOKS:

1. "Saibaba Reddy, E. Ramasastri, K. "Measurement of Engineering Properties of Soils", New age International (P) limited publishers, New Delhi, 2008.
2. G.VenkatappaRao andGoutham .K. Potable, "Geosynthetics Testing – A laboratory Manual", Sai Master Geoenvironmental Services Pvt. Ltd., 1st Edition 2008.
3. BrajaM.Das., "Soil Mechanics: Laboratory Manual", Oxford University Press, eighth edition, 2012

Online Learning Resources:

https://nptel.ac.in/courses/105101160